

Instruction Manual

MINIATURE CAPACITANCE POINT LEVEL SWITCH

FUNCTION:

Miniature Capacitance Level Switch comprises specially designed electronics and sensing probe using dedicated electronics and corresponding 2-Element Probe.

The unique feature of the instrument is the measured signal does not flow to the hopper / tank wall through the probe in the backward direction but detects actual media.

MEASURING SYSTEM:

The complete system consists of: -

The sensing probe and oscillator section along with embedded Electronic Switching unit

The sensing probe consists of two sections, (1) Measuring section, (2) ground section. Sensing section detects the change in dielectric strength & with the help of 100 KHz oscillator circuitry & given signal to electronic switching section.

PRIMARY AREA OF APPLICATION

Building industry materials, cement, sand, lime, etc

Foodstuff industry, water, milk powder, flour, salt, foodgrains, pharmaceutical

Automobile industry: oil, coolants

Plastic industry, powder, paints epoxy, resins ect

Timber industry, chemical and mining etc.

TECHNICAL SPECIFICATIONS:

SWITCHING UNIT:

Housing	SS, weatherproof, Customized
Optional installation	: Remote controller & probe / suitable of indoor-outdoor
Cable entry	: Customized
	Remote Electronics & probe/suitable for backpanel mounting.
Cable entry	3 nos. for Remote Type.
Ambient temperature	0°C to +125°C For Electronics
Power consumption	0.8 VA

Mains Voltage	12 VDC TO 36 VDC
Output	1 X SPDT Relay @ 5 Amps, Optional: PNP
Output	
Switching delay	Fixed 0.5sec. probe free / probe covered
Safety operation Order) (FSL/FSH)	Factory Setting: Hi/Low, ON /OFF (Specify before
Switch status display	Blue LED shows Normal and Red LED shows AlarmConditions.

SENSING PROBE:

Mounting :	Screwed 1/4" NPT / 3/4" NPT(Specify) Flanged (as per requirement)
Sense rod	Stainless steel
Insulation	PTFE (standard) / CERAMIC (Optional) / PEEK
Operating temperature	:125° C max.

INSTALLATION OF CAPACITANCE SWITCH :

Mounting of the probe would depend upon its construction and may differ from application to application. Depending upon the application, the rigid Probe can be mounted vertically, either from the top or horizontally, from the hopper side.

The standard unit has screwed, 1/2" BSP / NPT mounting, which can be mounted laterally on the container wall at the desired level of the material to be controlled. The probe should be horizontally or pointed slightly downward.

For installing the probe vertically from the top, sufficient clearance (equal to the probe length minimum) should be available above the hopper / Tank top to facilitate to hoist the probe over the hopper / tank top for insertion into the hopper / tank.

Following precautions should be taken during installations -The probe should not be bent or position distorted.

The Sense element of the probe (standard probe) should extend at least 50 mm into the vessel. The extended probe should be mounted in such a way that it does not extend further than necessary in the vessel.

During filling operation, the material should not fall directly onto the probe. Otherwise, protection guard should be provided over the probe.

During installation of probe with screwed mounting, turn the hexagonal mounting bush of the probe andnot the housing.

OPTIONAL REMOTE SWITCHING UNIT INSTALLATION:

The standard unit has electronics integral with the probe. For remote mounting of the Switching unit, Of the probe, the housing is suitable for back panel mounting.

For wiring and connection, refer the enclosed drawing. For remote unit only the supplied THREE TERMINAL connecting cable should be used for interconnecting probe with the Switching Unit. The Switching unit should not be mounted at the location where the ambient temperature is less than 60° Precaution should be taken to avoid fall of Sunrays on to the Switching Unit housing. In case it is not possible to avoid, a suitable Sun protection cover should be provided over the housing.

FAIL SAFE MODE:

Depending upon the process requirement, the minimum or maximum fail-safe mode can be defined capacitance level switch before to order processing

In Level Switch the Relay is in energized condition. When level changes state the relay de-energizes. Thus, besides level alarm condition, the operator gets an alarm even in case of mains failure or the instrument failure. This imparts a better overall reliability of operation.

Failsafe HI mode means the relay de-energizes when the level exceeds the desired or when mains supply fails.

Fail-safe LOW mode means the relay energizes when the level exceeds the desired level. NOTE: The contacts shown in the connection drawing are for Fail Safe Hi condition. The contacts will reverse when the FS is changed to Low position.

ELECTRICAL CONNECTION TO CAPACITANCE SWITCH:

Please refer the connection diagram for the electrical connection. Appropriate mains voltage should be connected to the terminals of the instruments as specified. The connectors are suitable for 1.5 sq.mm cable cross section.

SETPOINT SWITCHING CALIBRATION

There is single adjustment in the Switching Unit and is accessible from the top. There are two LEDs on top of the Switching Unit. The blue LED indicates the relay is in the energized state. The red LED indicates the alarm condition and relay in de-energized state. The setpoint adjustment is done by Blue variable resistor

(Please do not stress the Potentiometer, it is made of Plastic)

Turning the potentiometer clockwise will raise increase the sensitivity at which the relay operates and turning the potentiometer counterclockwise will lower the sensitivity at which relay operates.

Please use insulated tool (small screw driver with minimum metal portion in the insulated part) for adjustment. Do not turn the Pot beyond its limits, as it would damage the same.

Turn the potentiometer in the full clockwise position. Using the insulated tool and turn the potentiometer adjustment slowly to Counter clockwise until the relay just operates and Blue LED glows. Repeat the operation once or twice and note the point of switching.

Rotate the adjuster further ANTI-CLOCKWISE by 5%-10%, depending upon the application and the probe construction and mounting orientation. The instrument is set for desired switching point.

MAINTENANCE

In normal conditions the Capacitance SWITCH need no maintenance.

However, if the material has built up tendency, over a period of time, probe should be cleaned whenever needed occurs.

Ensure that the cable glands and the housing lid are sealed to prevent ingress of moisture.